

Building valid benchmarking datasets for cross-platform comparison

metagx

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How to build valid benchmarking datasets for cross-platform comparison

A comparison is only as good as the dataset behind it. If you want to see **real** comparative features between sequencing platforms (ONT vs Illumina vs PacBio), the dataset has to be built so that the *only* thing that differs is the variable you are studying. Everything else must be held constant. This note is the checklist for getting that right — written from the mistakes our own bundled datasets make (see the self-assessment at the end).

The one rule: *change one variable at a time*. To attribute a difference to “platform”, every other input — the genomes present, their relative abundance, and the sequencing depth — must be identical across the datasets you compare. Otherwise your “platform effect” is actually a depth effect or an abundance-model effect wearing a platform costume.

The three things that must match

1. SHARED REFERENCE AND KNOWN TRUTH

All datasets must derive from the **same reference genomes**, and you must record the ground truth so accuracy is measurable:

- the exact set of genomes (accessions) and their **lengths**,
- the **intended relative abundance** of each genome (the truth vector),
- provenance (which simulator/run produced each file).

Without a truth vector you can measure *agreement between platforms* but not *accuracy* (precision/recall, breadth, abundance error). Keep a `truth.tsv`: `accession, length_bp, rel_abundance`.

2. MATCHED ABUNDANCE MODEL — THE SUBTLE ONE

This is the trap that silently breaks most home-made benchmarks. Read simulators offer two fundamentally different notions of “abundance”, and they produce **different communities**:

Model	What it fixes	Effect	Tools
Equal coverage (depth ×)	reads per <i>base</i> per genome	read count $\propto$ genome length → long genomes dominate	wgsim, SimLoRD, ART (per-genome ×)
Equal reads (count)	reads per <i>genome</i>	even read share regardless of length	per-genome read-count sims

In our own data this produced a 12× distortion: under equal-coverage simulation the single 190 kb *Glossina* genome (44.5 % of total reference length) captured 44–60 % of reads, while under equal-reads simulation it captured ~8 %. **Same genomes, completely different profile.**

For a valid cross-platform comparison you must pick **one** abundance model and apply the **identical per-genome abundance vector** to every platform. If platform A is simulated equal-coverage and platform B equal-reads, you are comparing two different communities, not two platforms. (Either model is fine — equal-reads is more even and easier to reason about; equal-coverage mimics what real shotgun sequencing of equal-mass DNA actually does. Just be consistent and write down which.)

3. MATCHED SEQUENCING DEPTH

Depth must be comparable across platforms, normalized **the same way**:

- **Equal total bases** (Σ read length) — the fair choice for **assembly** comparisons, since assembly contiguity and breadth scale with bases of coverage, not read count.

- **Equal read count** — acceptable for **classification** comparisons (each read is one classification event), but unfair to short reads for assembly.

Mismatched depth is the most common confound. Our assembly experiment compared 20,000 / 8,000 / 415 / 415 reads and consequently measured *depth × platform*, not platform: PacBio HiFi at ~5× failed to assemble entirely, which says more about depth than about HiFi. Decide on a target (e.g. “30× per genome” or “20,000 reads”) and hit it on every platform.

Recipe: a clean simulated benchmark

```
# 0. Reference + truth -----
#   genomes.fasta = the shared community; write truth.tsv (accession,length,rel_abundance)

# 1. Pick ONE abundance vector and turn it into per-genome read counts -----
#   e.g. equal-reads: N_total / n_genomes reads each, OR a deliberate skew you record.
#   For equal-coverage, per-genome reads = depth * genome_len / mean_read_len.

# 2. Simulate EACH platform from the SAME per-genome read counts -----
#   Illumina : wgsim / ART      (short, ~150 bp, low error)
#   ONT      : pbsim3 / Badread (long, high error)
#   PacBio HiFi: pbsim3 multipass + ccs / SimLoRD multipass (long, low error)
#   PacBio CLR : pbsim3 1-pass / SimLoRD 1-pass (long, high error)
#   -> drive every simulator from the same per-genome counts so abundance is identical.

# 3. Normalize depth across platforms -----
seqtk sample -s42 platform.fastq <N_READS>      # equal read count, OR
#   downsample to equal total bases with a short script.

# 4. Record provenance -----
#   seed, simulator version, command, target depth, abundance model -> README per file.
```

For a quick smoke test you can subsample one deep FASTA into pseudo-replicates with `seqtk sample -s<seed>`, but pseudo-replicates share the same reads — fine for plumbing tests, not for measuring real biological or platform variance.

Better: a real multi-platform mock community

The strongest benchmark is one physical DNA sample sequenced on every platform — the abundance is then **intrinsically matched** (it is the same molecules), and any per-platform skew (GC bias, length bias, chimera rate) is a *genuine* platform feature you want to measure rather than a simulation artifact.

ZYMOBIOMICS — EXACT LINKS (THE DE-FACTO MULTI-PLATFORM MOCK)

Product info: ZymoBIOMICS Microbial Community Standard **D6300** (even, https://www.zymoresearch.com/products/zymobiomics-microbial-community-standardSMARTPANTS_PRESERVED_168) and Gut Microbiome Standard **D6331** (https://www.zymoresearch.com/products/zymobiomics-gut-microbiome-standardSMARTPANTS_PRESERVED_172).SMARTPANTS_PRESERVED_173

Platform	Community	Accession(s)	Where
Illumina + ONT (GridION/PromethION)	D6300 even / D6310 log	ENA study PRJEB29504 ; ONT FASTQ even ERR3152364 (GridION) / ERR3152365 (PromethION), log ERR3152366 / ERR3152367	https://lomanlab.github.io/mockcommunity/SMARTPANTS_PRESERVED_199 · https://www.ebi.ac.uk/ena/browser/view/PRJEB29504SMARTPANTS_PRESERVED_201SMARTPANTS_PRESERVED_202
PacBio HiFi	D6331 gut	NCBI PRJNA680590 : standard SRR13128014 , low SRR13128013 , ultra-low SRR13128012	https://github.com/PacificBiosciences/pb-metagenomics-tools/blob/master/docs/PacBio-Data.mdSMARTPANTS_PRESERVED_215SMARTPANTS_PRESERVED_216

Matched-community caveat: the public ONT + Illumina reads are the **D6300/D6310** standards; the public PacBio HiFi reads are the **D6331 gut** standard — *different communities*. So ZymoBIOMICS public data gives a clean **ONT-vs-Illumina** same-sample comparison (identical standards, Loman Lab), but a head-to-head that *includes* PacBio requires either sequencing one standard yourself on all three platforms, or treating the HiFi D6331 set as its own benchmark. This is exactly the “match the community” rule applied to real data.

Other options:

- **Loman Lab mock** (Nicholls 2019, ENA [PRJEB29504](https://www.ebi.ac.uk/ena/browser/view/PRJEB29504)) — ONT + Illumina of the same standards, with ground truth. The recommended *matched pair*.
- **CAMI / CAMI II** (data.cami-challenge.org) — simulated with gold-standard truth profiles and assemblies; abundance and depth are controlled by design.
- **Portik et al. 2022** (*BMC Bioinformatics* 23:541) — mock ONT + PacBio HiFi with profiling truth, purpose-built for long-read method benchmarking.

Validity checklist

- ☐ Same reference genomes across all platforms (record accessions + lengths).
- ☐ A written truth vector ([truth.tsv](#)) of intended per-genome relative abundance.
- ☐ **One** abundance model (equal-reads OR equal-coverage), applied identically to all.
- ☐ Depth normalized the **same way** for all platforms (equal bases for assembly; equal reads for classification) and at a level that actually supports the analysis (assembly needs $\geq \sim 10\text{--}30\times$ per genome; classification tolerates far less).
- ☐ Provenance recorded per file: simulator + version, seed, command, target depth.
- ☐ Analysis run identically across platforms (same DB, same params, same pipeline), varying only the platform-routed steps (QC, assembler, minimap2 preset).
- ☐ Confounds named in the writeup. If something is not matched, say so and read results qualitatively.

Self-assessment of the bundled metagx datasets

Honest accounting of what our shipped test data does and doesn't support:

Property	ONT (simulated_metagenomic_reads.-fasta)	Illumina (wgsim)	PacBio (SimLoRD)
Shared reference (30 genomes)	✓	✓	✓
Abundance model	equal-reads (even)	equal-coverage (length-biased)	equal-coverage
Depth	~20,000 reads	~8,000 reads	~415 reads (~5x)

Good for: [exercising each platform's pipeline path end-to-end \(QC routing, assembler routing, classification, the comparison machinery\)](#). This is their purpose and they do it well.

Not valid for, as-is: [a head-to-head platform accuracy ranking](#) — the ONT set uses a different abundance model than the uniform-coverage sets, and depths differ by ~48x. Experiments 07–08 therefore report results **qualitatively** and label the [con-founds explicitly](#).

To turn them into a valid benchmark: [re-simulate all four platforms from one per-genome abundance vector at one matched depth target \(e.g. 20,000 reads each, equal-reads model\), then re-run `metagx compare`](#). That single change converts the suite from “pipeline smoke tests” into “platform comparison with measurable accuracy.”

See also

- [DATASETS.md](#) — dataset catalog + experiments 07 (classification) and 08 (assembly).
- [scripts/compare_platforms.py](#) / [metagx compare](#) — the comparison engine.
- [results/experiments/cross_technology_comparison/](#) and [.../cross_platform_comparison/](#) — the comparisons that surfaced these lessons.